

1. The **Query** (Q) and **Key** (K) matrices are used together to determine the raw attention scores.

2. The raw scores obtained from the dot products are typically **scaled** down.
This scaling helps in stabilizing the gradients during training.
This is often done by dividing the scores by the square root of the dimension of the Key (K) vector.

3. The scaled scores are then passed through a **softmax** function, which turns them into probabilities.

These probabilities determine how much each element in the sequence should attend to every other element.

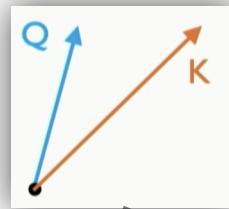
... and this results in the attention matrix

1. Encoding the **position**

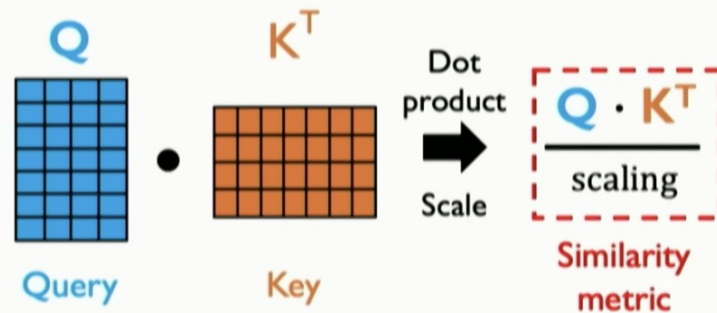
2. Extracting **query, key, value**

3. Compute **attention weighting**

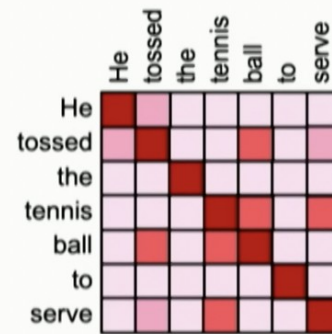
4. Extract **features with high attention**



Attention score: compute pairwise similarity between each **query** and **key**



Attention weighting: where to attend to!
How similar is the key to the query?



$$\text{softmax} \left(\frac{Q \cdot K^T}{\text{scaling}} \right)$$

Attention weighting